Higher National Diploma in Engineering (Mechanical/Civil)

Higher National Diploma in Building Service Engineering

First Year, First Semester Examination – 2019

CE-1102-Fluid Mechanics

Answer any five questions. **3hours**

Question 1 (Q1)

a. Show that the magnitude of the Hydrostatic thrust F on a plate is given by:

$$F = AP_G$$

Where $A =$ Submerged area and $P_G =$ pressure at the centroid of the submerged surface. (Marks 6)

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b. The gate in **Fig. 1** is $5\ \mathrm{m}$ wide, is hinged at point **B**, and rests against a smooth wall at point **A**. Compute, i. the force on the gate due to the water pressure ii. the horizontal force P exerted by the wall at point A iii. the reactions at hinge B. (Marks 14)

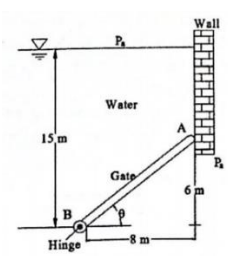

Fig.1

Question 2 (Q2)

a. Explain the following terms. i. Viscosity ii. Newtonian and Non-Newtonian fluids iii. "Newton's law" with respect to viscosity (Marks 06)

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b. The rotating cylinder viscometer shears the fluid in a narrow clearance, t as shown in Fig. 2. Assume a linear velocity distribution in the gaps. If the driving torque T is measured, find an expression for μ by i. neglecting, and ii. including the bottom friction. (14 Marks)

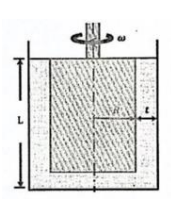

Fig.2

Question 3 (Q3)

a. Explain the losses that could be expected in pipe flow (06 Marks)

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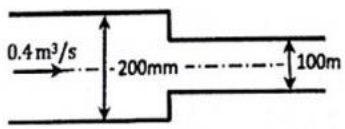
b. A $\mathbf{200\ mm}$ diameter pipe reduces suddenly in diameter to $\mathbf{100\ mm}$, as shown in Fig. 3. The pipe is carrying water at the rate of $\mathbf{0.4\ m^3/s}$ and the coefficient of contraction is $\mathbf{0.6}$. Calculate: i. the head loss due to contraction and ii. the pressure difference between the two sections (14 Marks)



Question 4 (Q4)

a. Explain the following equations mentioning the assumption made in deriving them. i. Continuity equation ii. Bernoulli's equation (04 Marks)

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b. Starting with the Continuity and Bernoulli equations derive the expression to measure fluid flow rate for the Fig. 4 shown below. (06 Marks)

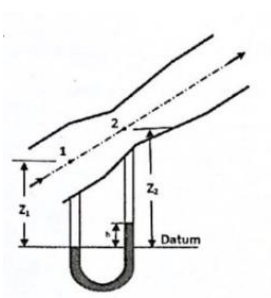

Fig. 4

c. Also show that when the pressure difference is measured using a mercury manometer the following expression can be used:

$$Q_{\text{Actual}} = C_d A_1 A_2 \sqrt{\frac{2gh\left(\frac{\rho_m}{\rho} - 1\right)}{{A_1}^2 - {A_2}^2}}$$

Where C_d is the coefficient of contraction, A_1 and A_2 are the cross-sectional areas at elevations Z_1 and Z_2 respectively, ρ_m and ρ are the densities of the mercury and fluid respectively, and h is the manometer head difference. (10 Marks)

Question 5 (Q5)

a. Explain the forces due to the flow around a pipe bend (05 Marks)

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b. A pipe bend has a cross-sectional area of $\mathbf{0.01\ m^2}$ at inlet and $\mathbf{0.0025\ m^2}$ at outlet. It bends $\mathbf{90^\circ}$ from its initial direction in a horizontal plane. The velocity of flow of water is $\mathbf{4\ m/s}$ at inlet with a pressure of $\mathbf{100\ kPa}$ gauge. Draw the diagram and calculate the force acting on the bend parallel and perpendicular to the initial direction. (15 Marks)

Question 6 (Q6)

a. Water flows through a $\mathbf{2\ cm}$ diameter pipe at $\mathbf{1.6\ m/s}$. Calculate the **Reynolds number (Re)**. Also find the velocity required to give the same Reynolds number when the pipe is transporting air. (08 Marks)

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b. Assuming the pressure loss along a pipe Δp, can be expressed in terms of the following:

- fluid density ρ
- kinematic viscosity ν
- diameter d
- velocity u Show that the pressure loss can be expressed as:

$$p = \rho u^2 \varphi(\mathrm{Re})$$

Hence find the ratio of pressure drops in the same length of pipe for both cases.

You will need to use these physical properties:

Variable	Water	Air
ρ	1000 kg/m³	1.19 kg/m³
ν	1.31×10^{-6} m²/s	15.1×10^{-6} m²/s